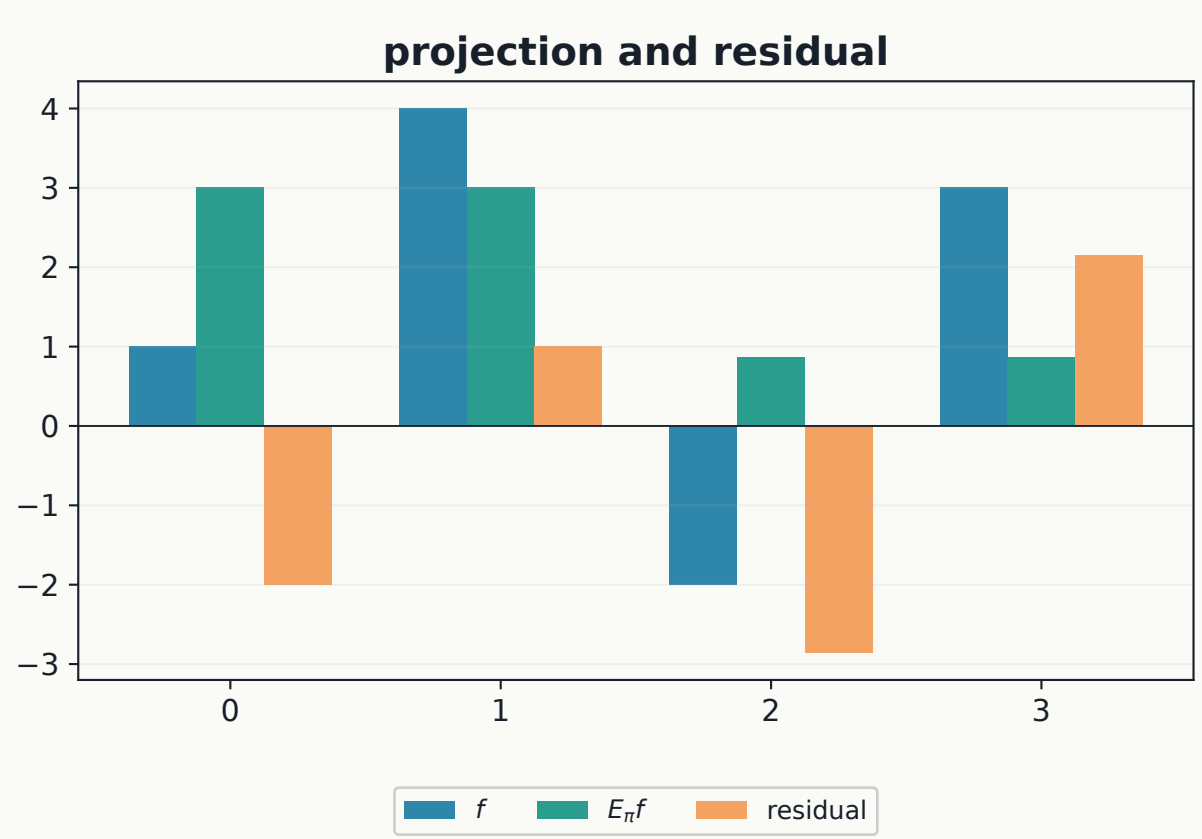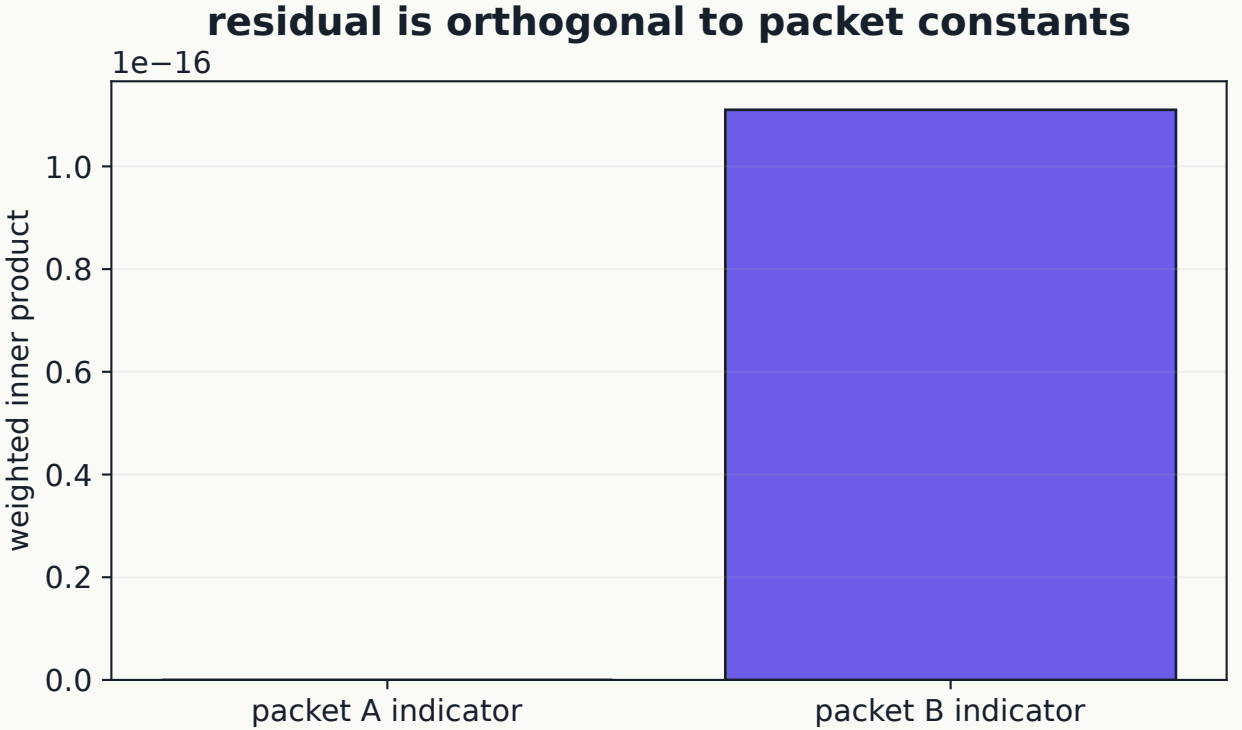Conditional expectation is a genuine weighted orthogonal projector